

Weekly Report — 2026-07-14 → 2026-07-21 (all workstreams)

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Scope: Full work log since the v2.1 report (2026-07-13), across both agent lanes and platform work: the OBD lane's expert-review cycle was closed and its golden tier locked (15/15, mean 0.938); the manual lane went v2.1 → v2.2 (0.777 → 0.831, first all-green run) through three root-cause fixes; two production deployments shipped; and the repo's report/eval archive was brought fully up to date.

1. Executive summary

Lane	Milestone this week
OBD agent	Expert-review cycle CLOSED: 2 goldens revised per reviewer comments, post-revision eval 15/15 pass holds (mean 0.938), all 15 promoted into the locked tier. The OBD eval now runs on expert-approved goldens by default.
Manual agent	v2.2 = 0.831 mean, 30/30 above the pass floor (first all-green full run; v2.1 was 0.777). Three root-cause fixes: DTC-index navigation, NOTE-banner section slicing, deterministic vehicle injection.
Production	Both manual-lane fixes deployed to the PolyU server (2 single-container redeploys, warm LLM preserved throughout — zero cold-load windows).
Repo hygiene	5 feature PRs + 2 archive PRs merged; 4 issues closed (#204, #205, #210, #213); all report deliverables (06-27, 07-13, 07-21) and milestone eval artifacts now tracked in-repo; single-main branch state restored on GitHub / local / server.

2. OBD agent lane — HARNESS-27: expert-review closure (07-19 → 07-20)

The 15 OBD goldens had been awaiting expert re-review since 2026-06-08, when reviewer Deng Xiao left 2 of 15 entries at needs_revision. This week closed the loop end to end:

a) Golden revisions per review comments (#205 / PR #206)

- compound-001 — reviewer: "idle RPM is not 0; state the basis of the no-fault claim." Fixture re-analysis confirmed RPM=0 only for the first 3 samples (engine off) and true idle at ~1475–1990 rpm (median ~1550). The summary now separates engine-off from idle, spells out the multi-signal basis for the no-misfire/no-overheat conclusion, and adds a caveat that 1 Hz sampling cannot exclude brief misfire.
- dtc-decode-002 — reviewer: per the Yamaha manual (pp. 8-36/8-37) P0117 is "ECT sensor — short to ground detected", not just the generic SAE "circuit low voltage". The summary now gives both definitions with page references, and the pitfall accepts either phrasing.

b) Post-revision eval re-run (#205 / PR #207)

Full 15-golden run on the shipping qwen3.5:27b-q8_0 (28m51s): 15/15 pass holds, mean 0.938 (harness27_obd_rerun.json, supersedes harness25_obd_round1.json as the OBD gate reference). Both revised entries hold (compound-001 0.839 → 0.840, dtc-decode-002 0.800 → 0.800). One live judge failure (the #201 "empty judge content" mode) hit compound-001 mid-run and was resolved by a standalone re-run merged into the report with provenance. Sole notable delta: adversarial-002 0.993 → 0.742 (phrasing variance, judge itself scored 0.95).

c) Locked-tier promotion (#205 / PR #208)

All 15 goldens promoted into golden/v2/locked/yamaha_road_test.jsonl via promote_golden.py against the prod review DB: 13 through the normal review-quality gate (5-star accepts), the 2 revised entries with --force + out-of-band re-approval, all recorded in PROMOTIONS.md. This closes the HARNESS-20 safety-net gap (the locked OBD file had shipped empty) — both eval lanes now run entirely on expert-approved locked goldens by default. Issue #205 closed.

3. Manual agent lane — v2.1 → v2.2 (07-19 → 07-21)

Headline

Milestone	Mean (30 locked goldens)	Notes
v2.1 (2026-07-13)	0.777	Median 0.811, σ 0.195.
v2.2 (2026-07-21)	0.831	Median 0.887, σ 0.136, 30/30 above the pass floor — first all-green full run. Mean wall 72 s.

Spread tightened by a third — the gain came from fixing the worst entries, not inflating the top. Per question type:

Type	v2.1	v2.2
lookup	0.839	0.841
procedural	0.741	0.945
cross-section	0.752	0.792
image-required	0.779	0.815
adversarial	0.774	0.759

Procedural — the weakest category at v2 (0.567) — is now the strongest.

Work items

Work item	Issue / PR	Measured effect
Judge blind to figures the agent surfaced (held last week pending confound resolution)	#193 / PR #197 (merged 07-19)	image-required 0.686 → 0.785 in the same-rubric re-run
Re-measure the lane after the WP1/WP3 prompt fixes (numbers were 9 days stale)	re-run 07-21	0.705 → 0.776 (22/30); confirmed WP1/WP3 worked and isolated 2 structural failures
DTC Quick Index mapped codes to nothing (occurrence counts only); P-code sections depth-4 invisible with parent chain cut by junk NOTE banners	#210 / PR #211 (HARNESS-28)	dtc-001 0.400 → 0.988; procedural-002 0.363 → 0.985; P0107 section content recovered 112 → 15,264 chars
Vehicle identity reached the manual sub-agent only via LLM free text — vehicle-less inquiries made manual selection a coin flip (cross-004 answered a Yamaha question from the Corolla Haynes manual)	#213 / PR #214 (HARNESS-29, closes #204)	Deterministic VEHICLE block injected from the session row (APP-60 make/model + VIN); cross-004 0.347 → 0.755, zero wrong-manual reads in all verification runs
v2.2 milestone baseline	PR #215	0.831, 30/30 all-green (harness29_v2_2_baseline.json)

Design decision of note (HARNESS-29): vehicle identity is injected by harness code, not LLM text — extending the APP-59/APP-60 "identity is captured at upload" guarantee through the delegation link. The eval mirrors the exact production interface (GoldenEntry.vehicle + corpus default), so no golden files changed and no locked-tier re-review was triggered.

4. Platform, deployment & repo hygiene

- Two production deployments (PolyU server) using the single-container recreate procedure — Postgres and the warm Ollama model stayed up through both; no cold-load window for users. Prod now runs full v2.2 code.
- Boundary items merged 07-13 after the v2.1 report was issued: APP-64 (PR #189) — V1 premium diagnose now passes the session's vehicle_model to retrieval (same vehicle-identity thread as HARNESS-29); HARNESS-25 OBD round-1 report (PR #190).
- v2.1 fix-stack artifacts landed 07-14 (#196 coverage gate + manual pinning, #198 caption-stub demotion, #199 slug-tolerant claim_precision): the code was covered in last week's report; the 53-min v2.1 milestone eval it produced (phase6_v2_1_manual_lane.json) had been left untracked and was committed retroactively this week (PR #209) with provenance.

- Report archive completed: 06-27 progress report and 07-13 weekly report (md+pdf) — previously existing only on local disk — are now tracked under docs/reports/, alongside this report.
- Branch state: all feature branches deleted after merge on GitHub, local, and server; final state is a single main everywhere; 0 open PRs at week close.
- Eval-run artifacts committed this week under docs/eval-reports/: harness27_obd_rerun.json, harness24_manual_rerun_20260721.json, harness28_manual_baseline_20260721.json + harness28_dtc_slug_targeted.json + harness28_dtc_slug_round2.json, harness29_vehicle_targeted.json, harness29_v2_2_baseline.json, phase6_v2_1_manual_lane.json.

5. Honest caveats

- Manual lane: the 30/30 all-green is against the 0.4 regression floor; the 0.7 quality bar stands at 24/30 (both week-over-week highs). Run variance is reduced but real (cross-003 swung 0.802 → 0.425 → 0.745 across three same-day runs). Four stable low performers are now clearly isolated: lookup-005 (0.578 ×4 runs), image-006 (0.612 ×3), cross-006 (0.483–0.612), adversarial-006 (0.646–0.660).
- OBD lane: dtc-decode-002 sits exactly at the 0.7 threshold (0.800 overall is healthy; its weakest rubric line is the decode pivot) — the lane's most fragile entry. adversarial-002's 0.993 → 0.742 swing is the OBD lane's variance analogue; same caution as the manual lane: trust means, not single entries.
- cross-004's residual gap (0.755 vs 0.95+) is the spec-lookup half of its two-part question — manual selection is fixed, cross-section spec retrieval is not.

6. Next steps

1. Root-cause the two zero-variance manual-lane lows: lookup-005 and image-006 (score stability across runs suggests structural causes, like the DTC-navigation class fixed this week).
2. Characterise cross-006 and adversarial-006 (stable-low vs variance).
3. cross-004 spec-lookup completion (second read into the engine-specs table).
4. OBD lane enters maintenance: harness27_obd_rerun.json is the regression gate; next full run only after agent-affecting changes.
5. Umbrella #116 continues: next milestone eval after the above lands becomes v2.3.